Quiz ~~9~~ 10/In-class Assignment

Name: ____________________________ PSID: __________

Problem 1

7.43 At $\omega = 400$ rad/s, the input impedance of the circuit in **Fig. P7.43** is $\mathbf{Z} = (74 + j72)\ \Omega$. What is the value of L?

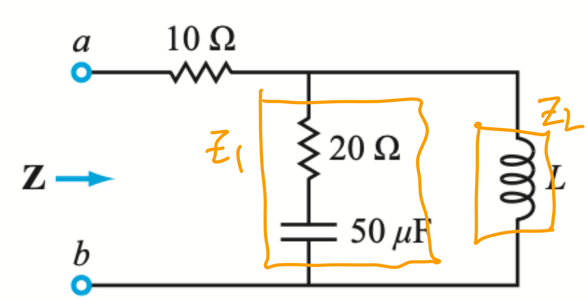


Figure P7.43 Circuit for Problem 7.43.

$$Z_1 = 20[\Omega] - \frac{j}{400[\frac{rad}{s}]\cdot 50[\mu F]} = (20 - j50)[\Omega]$$

$$Z = 10[\Omega] + Z_1 \| Z_L = (74 + j72)[\Omega]$$

$$Z_1 \| Z_L = \frac{Z_1 Z_L}{Z_1 + Z_L} = (64 + j72)[\Omega]$$

$$Z_1 Z_L = (64 + j72)[\Omega](Z_1 + Z_L)$$

$$Z_L\left(Z_1 - (64 + j72)[\Omega]\right) = (64 + j72)[\Omega]\cdot Z_1$$

$$Z_L = \frac{(64 + j72)(20 - j50)}{20 - j50 - (64 + j72)}[\Omega]$$

$$Z_L = j40[\Omega]$$

$$Z_L = j\omega L$$

$$L = \frac{Z_L}{j\omega} = \frac{j40[\Omega]}{j\,400[\frac{rad}{s}]}$$

$$L = 0.1[H]$$

$Z_L =$ $j40[\Omega]$

$L =$ $100[mH]$

Problem 2

7.49 The circuit in **Fig. P7.49** is in the phasor domain. Determine its Thévenin equivalent at terminals (a, b).

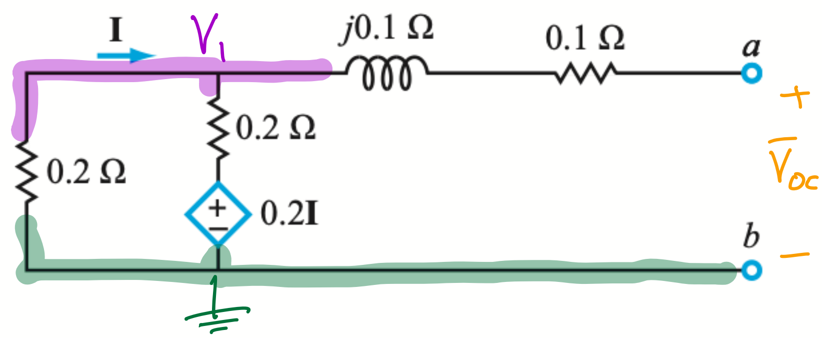


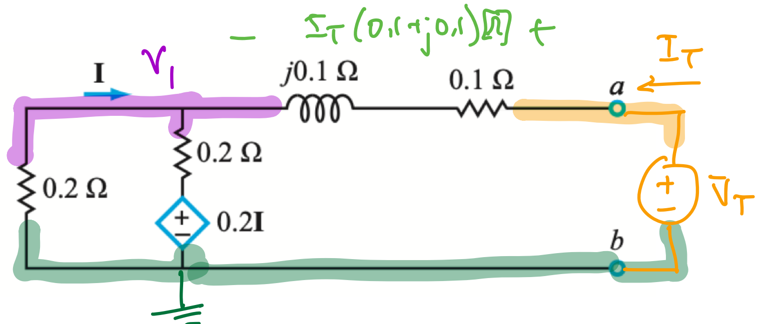
Figure P7.49 Circuit for Problem 7.49.

$$\left[\frac{V_1 - 0.2[\Omega]I}{0.2[\Omega]} + \frac{V_1}{0.2[\Omega]} = 0\right] \times 0.2[\Omega] \qquad I = -\frac{V_1}{0.2[\Omega]}$$

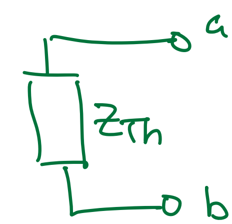
$$2V_1 + V_1 = 0$$

$$V_1 = 0$$

this makes sense – no independent source to activate the dependent source



Thevenin Equiv.:



$$\left[\frac{V_1}{0.2[\Omega]} + \frac{V_1 - 0.2[\Omega]I}{0.2[\Omega]} - I_T = 0\right] \times 0.2[\Omega]$$

$$\overbrace{V_1 + V_1 - 0.2[\Omega]\left(-\frac{V_1}{0.2[\Omega]}\right)}^{3V_1} - 0.2[\Omega]I_T = 0 \Rightarrow 3V_T - I_T(0.3+j0.3)[\Omega] - 0.2[\Omega]I_T = 0$$

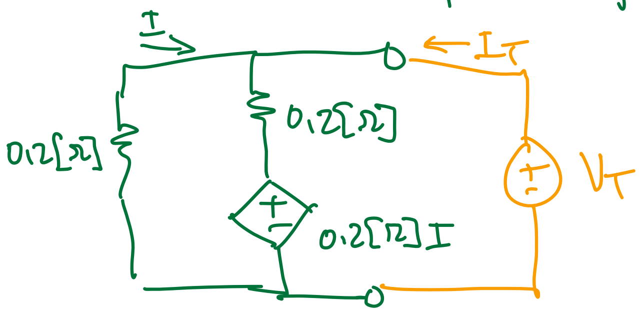
$$3V_T = (0.5+j0.3)[\Omega]\, I_T$$

$$V_T = V_1 + I_T(0.1+j0.1)[\Omega]$$

$$V_1 = V_T - I_T(0.1+j0.1)[\Omega]$$

$$Z_{Th} = \frac{V_T}{I_T} = \frac{(0.5+j0.3)[\Omega]}{3}$$

you can get to the answer faster by first finding the equiv. resistance of the left hand side of the circuit:



$$V_T = -I \cdot 0.2[\Omega] \Rightarrow I = \frac{-V_T}{0.2[\Omega]}$$

$$(I + I_T)\,0.2[\Omega] + 0.2[\Omega]I = V_T$$

$$\left(-\frac{V_T}{0.2[\Omega]} + I_T\right)0.2[\Omega] - 0.2[\Omega] \cdot \frac{V_T}{0.2[\Omega]} = V_T$$

$$-V_T + 0.2[\Omega]I_T - V_T = V_T$$

$$3V_T = 0.2[\Omega]\,I_T$$

$$\frac{V_T}{I_T} = \frac{0.2[\Omega]}{3}$$

$$Z_{Th} = \frac{V_T}{I_T} + (0.1 + j0.1)[\Omega]$$

$$= \frac{0.2[\Omega]}{3} + \frac{0.3 + j0.3}{3}[\Omega]$$

$$Z_{Th} = \frac{0.5 + j0.3}{3}[\Omega]$$